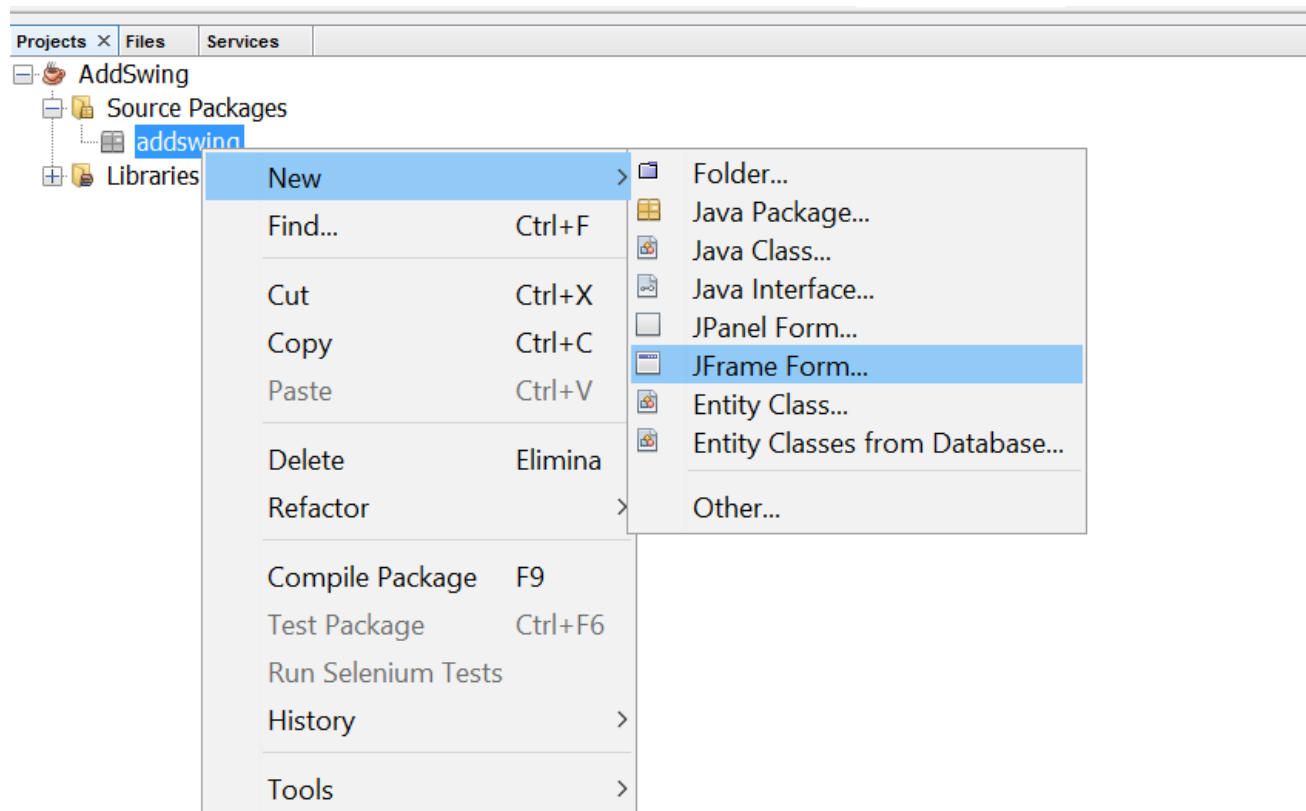


La prima applicazione SWING

- Aggiungere un nuovo JFrame



La prima applicazione SWING

The screenshot shows the NetBeans IDE 8.1 interface for creating a Swing application. The main window is titled "AddSwing - NetBeans IDE 8.1". The menu bar includes File, Edit, View, Navigate, Source, Refactor, Run, Debug, Profile, Team, Tools, Window, and Help. The toolbar shows various icons for file operations and running the application. The left sidebar contains the "Projects" window, showing the project structure: "AddSwing" (Source Packages), "addswing" (Add.java), and "Libraries". The central area is the "Design" view, showing a blank canvas with a light gray background. The right sidebar contains the "Palette" window, which lists various Swing components under categories like "Swing Containers", "Swing Controls", "Swing Menus", "Swing Windows", "Swing Fillers", "AWT", "Beans", and "Java Persistence". The "Properties" window is also visible, showing the properties of the selected component (JFrame). The "Events" tab in the Properties window is selected, showing a list of events and their corresponding actions.

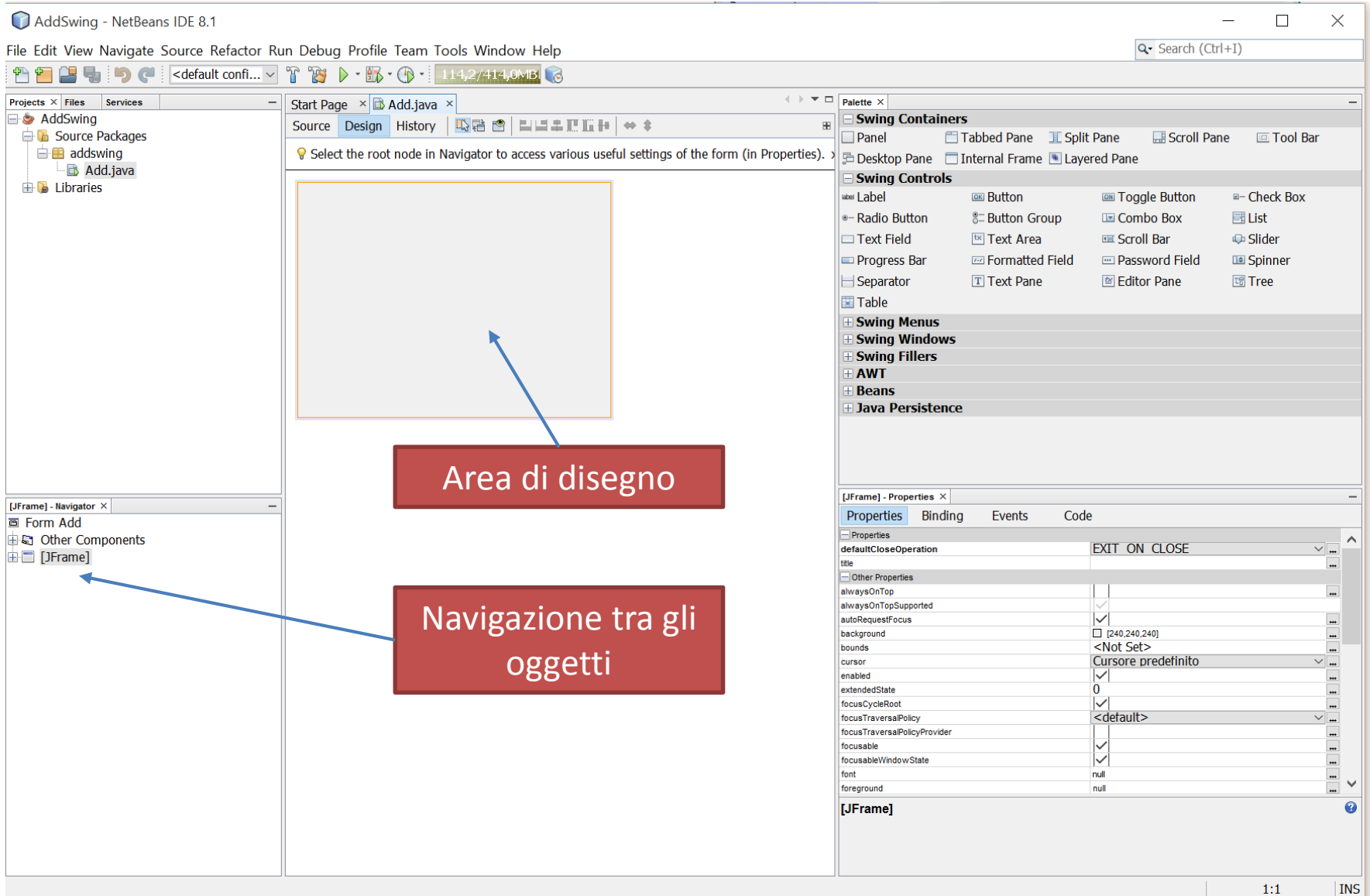
Paletta di componenti SWING

Proprietà degli oggetti

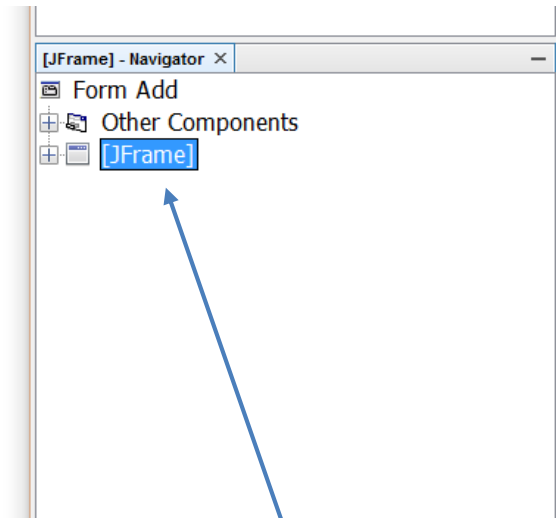
Eventi

Property	Value
defaultCloseOperation	EXIT ON CLOSE
title	
alwaysOnTop	
alwaysOnTopSupported	
autoRequestFocus	
background	[240,240,240]
bounds	<Not Set>
cursor	Cursore predefinito
enabled	
extendedState	0
focusCycleRoot	
focusTraversalPolicy	<default>
focusTraversalPolicyProvider	
focusable	
focusableWindowState	
font	null
foreground	null

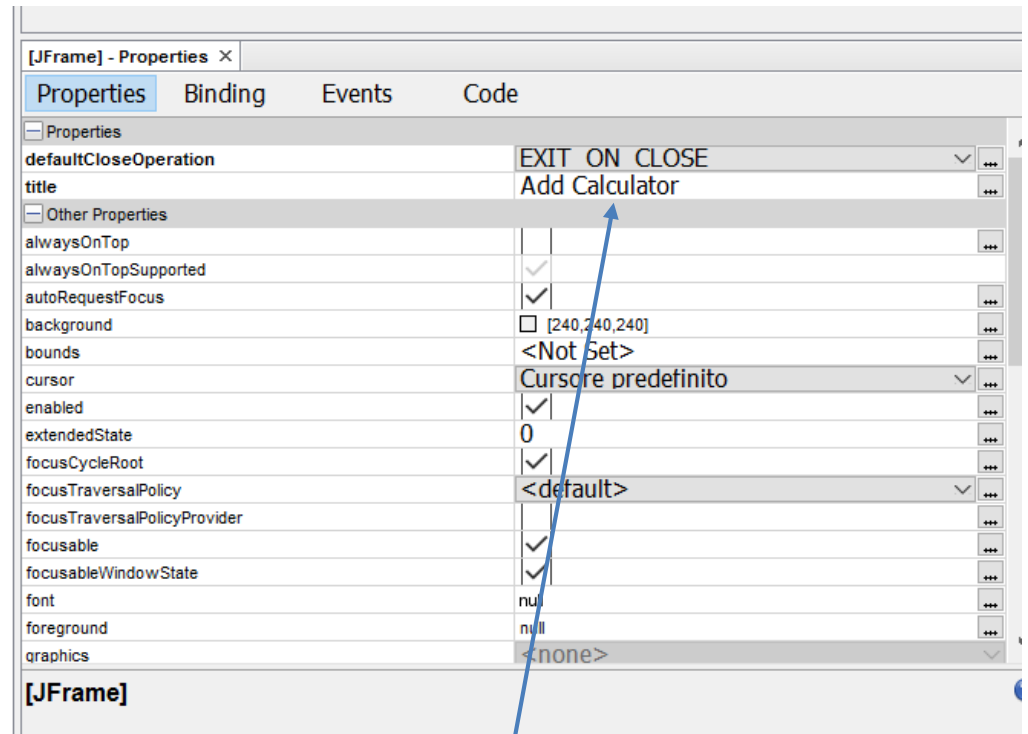
La prima applicazione SWING



La prima applicazione SWING



Seleziona il JFrame
tra i componenti



Modifica la proprietà
titolo (title)

La prima applicazione SWING

- Drag and drop delle componenti nell'area di disegno
- Selezionare le componenti dalla Palette
 - 2 JTextField per l'inserimento del testo (due operandi)
 - jtfOperand1, jtfOperand2
 - 1 JLabel per la stampa del risultato
 - jlResult
 - 1 JButton per effettuare la somma
 - jbAdd

La prima applicazione SWING

The screenshot shows the NetBeans IDE with the following components:

- Projects:** AddSwing, Source Packages, addswing, Add.java, Libraries.
- Source:** Add.java (Design mode).
- Design:** The Navigator window displays a tree hierarchy of components in the opened form. It shows a JFrame with two text fields (jtfOperand1 and jtfOperand2) and a button.
- Properties:** The Properties window for the selected text field (jtfOperand2) shows various attributes: editable (checked), background (255,255,255), columns (15), document (<default>), font (Tahoma 16 Plain), foreground (0,0,0), horizontalAlignment (LEADING), text (LEADING), tooltipText (LEADING), and UI (TextFieldUI).

Annotations and actions:

- TextField per gli operandi** (points to the text fields in the design view).
- Actions for text fields:**
 - cambiare la dimensione
 - rimuovere il testo
- Actions for button:**
 - cambiare la dimensione
 - rimuovere il testo

La prima applicazione SWING

The screenshot displays the NetBeans IDE environment for creating a Java Swing application. The main window, titled 'Add.java', is in the 'Design' tab, showing a visual representation of the application. It features two text input fields at the top, a 'Result:' label, and an 'Add' button below it. A red rectangular box highlights the 'Add' button, and a blue arrow points from a red text box to it. The 'Palette' on the right lists various Swing components, including containers (Panel, Desktop Pane, etc.) and controls (Label, Button, Text Field, etc.). The 'Form Add' window at the bottom left shows a list of components available for the form, including 'jtfOperand1 [JTextField]', 'jtfOperand2 [JTextField]', 'jlResult [JLabel]', and 'jbAdd [JButton]'. A red rectangular box highlights the 'jbAdd [JButton]' component, and a blue arrow points from the same red text box to it.

Le componenti presenti nel JForm

La prima applicazione SWING

- La gestione degli eventi
 - quando il bottone (jbAdd) viene premuto l'applicazione deve calcolare la somma
 - leggere i valori dai JTextField
 - sommare i valori
 - modificare il testo della JLabel con il risultato
 - selezionare il componente (jbAdd) e il tab «Events» dalle proprietà dell'oggetto

La prima applicazione SWING

- Gestione dell'evento

The screenshot illustrates the process of setting an event listener for a JButton in the NetBeans IDE. The 'Form Add' window contains several components: jtfOperand1 [JTextField], jtfOperand2 [JTextField], jlResult [JLabel], and jbAdd [JButton]. A context menu is open over jbAdd, showing the 'Events' tab selected, which has opened a submenu with 'Action' selected, and 'actionPerformed' selected. A blue arrow points from the 'actionPerformed' selection to the 'Events' tab in the 'jbAdd [JButton] - Properties' window.

Properties	Binding	Events	Code
Events			
actionPerformed			<none>
ancestorAdded			<none>
ancestorMoved			<none>
ancestorMoved			<none>
ancestorRemoved			<none>
ancestorResized			<none>
caretPositionChanged			<none>
componentAdded			<none>
componentHidden			<none>
componentMoved			<none>
componentRemoved			<none>
componentResized			<none>
componentShown			<none>
focusGained			<none>
focusLost			<none>

```
23      * WARNING: Do NOT modify this code. The content of this method is always
24      * regenerated by the Form Editor.
25      */
26      @SuppressWarnings("unchecked")
27      Generated Code
80
81      private void jbAddActionPerformed(java.awt.event.ActionEvent evt) {
82          // TODO add your handling code here:
83      }
84
```

La prima applicazione SWING

```
private void jbAddActionPerformed(java.awt.event.ActionEvent evt) {  
    try {  
        int op1 = Integer.parseInt(jtfOperand1.getText());  
        int op2 = Integer.parseInt(jtfOperand2.getText());  
        jlResult.setText("Results: " + (op1 + op2));  
    } catch (NumberFormatException nfex) {  
        JOptionPane.showMessageDialog(this, "No valid number " + nfex.getMessage(),  
            "Error", JOptionPane.ERROR_MESSAGE);  
    }  
}
```

- Lettura del testo dalle *JTextField* con il metodo *getText*
- Conversione delle stringhe in interi
- Stampa del risultato
- Blocco *try-catch* per catturare eventuali errori di formato degli interi
 - visualizzazione di un messaggio di errore tramite *JOptionPane*

Eseguire l'applicazione

